LoRA-QA: Parameter-Efficient Fine-Tuning of Sub-8B Models for Question Answering

Mei Tanaka

Kestrel AI Lab

Workshop on Efficient Language Models (WELM), 2024

Abstract

LoRA-QA is a recipe for adapting a Small Language Model to Open-domain QA with Low-Rank Adaptation. The method starts from TinyDistill checkpoints and trains low-rank adapters on a single consumer GPU. On Natural Questions and HotpotQA, 1.3B, 3B, and 7B models approach full fine-tuning while updating about 1% of parameters. The method reduces memory and training time without changing the base model, making repeated task adaptation practical for small teams.

Keywords

Parameter-efficient fine-tuning, Low-Rank Adaptation, Natural Questions, HotpotQA

1. Introduction

Fine-tuning a Small Language Model can still be costly when every parameter is updated. LoRA-QA uses Low-Rank Adaptation to constrain updates and begins from TinyDistill student checkpoints. The evaluation covers Open-domain QA on Natural Questions and HotpotQA, with full fine-tuning as the generic comparison. This paper builds directly on TinyDistill [1].

2. Method

For each transformer projection selected by the recipe, LoRA-QA adds a pair of low-rank matrices while freezing the original weights. Rank is chosen per layer from a small schedule, with higher rank in attention output projections and lower rank elsewhere. TinyDistill supplies the initial checkpoint, and training examples are mixed across Natural Questions and HotpotQA. Gradient accumulation keeps the run within one consumer GPU memory budget. We report Exact Match, trainable-parameter fraction, and GPU hours for 1.3B, 3B, and 7B models. Adapters are saved separately, so multiple task variants can share the same base checkpoint.

3. Experiments

LoRA-QA stays within one EM point of full fine-tuning on all reported settings while training 1.0% of parameters. On Natural Questions, the 3B adapter reaches 55.8 EM versus 56.4 for full fine-tuning; on HotpotQA it reaches 34.1 versus 34.8. GPU hours are 3.2, 6.7, and 14.9 for 1.3B, 3B, and 7B models.

Model	NQ EM	HotpotQA EM	GPU hours
1.3B	51.0	29.5	3.2
3B	55.8	34.1	6.7
7B	60.2	39.0	14.9

4. Conclusion

LoRA-QA makes Small Language Model adaptation practical with Low-Rank Adaptation and TinyDistill initialization. It preserves nearly all full fine-tuning accuracy while sharply reducing trainable parameters and GPU hours. Future work should study adapter composition across more domains.

References

[1] Tanaka, M. (2023). TinyDistill: Knowledge Distillation into Compact Language Models. Workshop on Efficient Language Models (WELM).